

Spaced Repetition Enhances Self-Rated Learning Confidence

A Large Randomized Trial Among Practicing Family Physicians

Ting Wang, PhD • Zachary J. Morgan, MS • Andrew Bazemore, MD, MPH • Warren P. Newton, MD, MPH • David W. Price, MD

J Contin Educ Health Prof. 2026;46(1):16–22

Presented by Ting Wang, PhD, American Board of Pediatrics | ABMS Conference, September 2026

Why Metacognitive Accuracy Matters

Confidence that matches correctness is a clinical competency

Metacognitive accuracy

The degree to which a clinician's confidence in a judgment matches its actual correctness.

Helps learners know what they know, identify gaps, and regulate their own learning.

Overconfidence

Confidence exceeds correctness.

A recognized contributor to diagnostic error (Berner & Graber, 2008).

Underconfidence

Correctness exceeds confidence.

Delays critical decisions, prompts unnecessary testing, disrupts workflow, and raises cost of care.

Why it matters for lifelong learning

Metacognitive skill supports movement up Miller's pyramid (knows → shows how → does) and across Moore's outcome levels 3–5 (learning → competence → performance). Practicing clinicians must continually recalibrate confidence as clinical knowledge evolves.

Sources: Pintrich 2002; Fleming 2024; Berner & Graber 2008; Miller 1990; Moore et al. 2009.

What We Know — and What We Don't

Established evidence

- Clinicians consistently misjudge their own competence; self-assessment is a cognitive-bias problem (Eva & Regehr 2005; Davis 2006).
- Spaced repetition outperforms massed study for long-term retention and transfer across medicine, nursing, and psychology (Kerfoot 2007; Cepeda 2006).
- In the ABFM longitudinal assessment, spaced repetition improved learning and knowledge transfer (Price et al. 2025).
- Post-item confidence ratings are strongly associated with sex and age, so raw ratings are not a clean learning signal.

The gap

- Few studies examine how repetition strategies change clinicians' confidence, i.e., metacognitive accuracy rather than accuracy alone.
- Unknown whether there is a dose–response relationship between number of repetitions and confidence gain.
- Unknown whether confidence gains transfer to novel (clone) items that test the same point in a new clinical scenario.

Research question

How does self-rated confidence change under different spaced repetition strategies, for repeated and for clone questions, in a large randomized cohort of practicing family physicians?

Instrument: ABFM Knowledge Self-Assessment (CKSA)

A low-stakes, longitudinal formative assessment used in continuing certification

25

multiple-choice questions delivered each quarter

Web + mobile

platform, completed at the physician's own pace

1 rating

of confidence immediately after every answer

6-point confidence scale



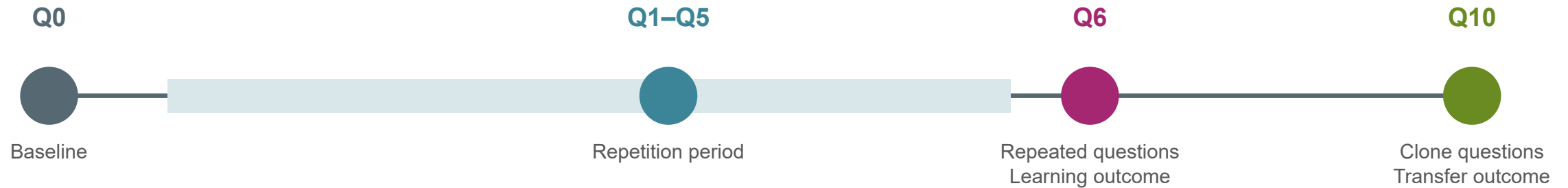
Not at all confident

Extremely confident

- Even number of points, no midpoint: designed to avoid central-tendency bias.
- Grounded in established principles of metacognitive assessment; prior work on this platform shows face validity and discrimination across confidence levels.
- Treated as continuous in analysis (ordinal, more than five ordered categories; Norman 2010).
- Not externally validated in this study (acknowledged limitation).

Study Design: Randomized Spaced Repetition Trial

Six arms, five calendar quarters of repetition, outcomes measured at quarter 6 and quarter 10



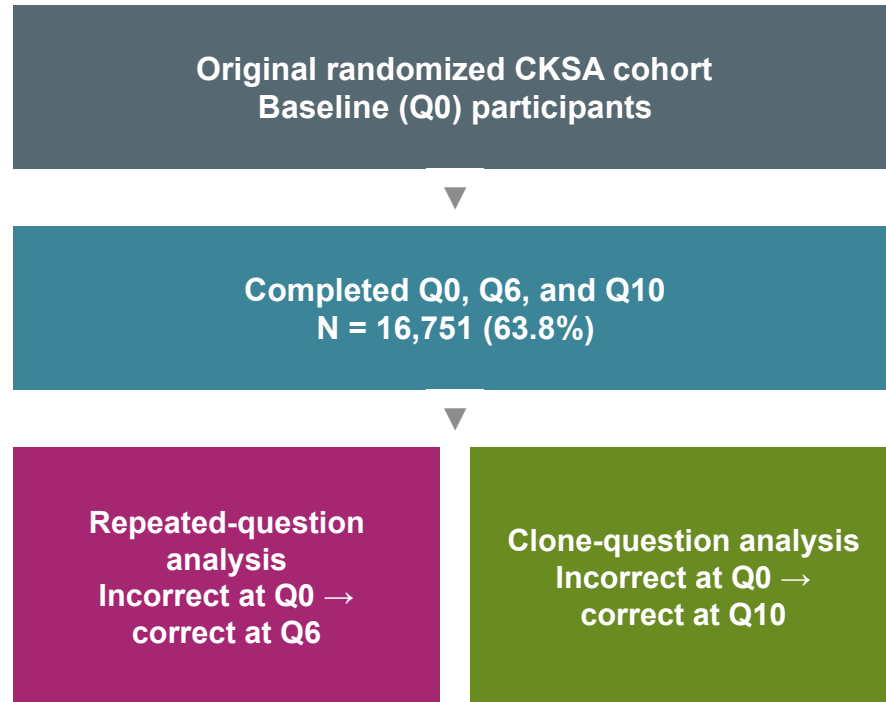
Control	Groups 1–2	Groups 3–5
n = 2,896 No repeated questions during the six quarters	n = 2,831 / 2,744 One repetition of selected questions, at differing intervals	n = 2,777 / 2,755 / 2,748 Two repetitions of selected questions, at differing intervals

Clone question: a rewritten version of an original item that preserves the testing point but changes patient characteristics, distractors, or numerical values. Randomization at the participant level; approved by the American Academy of Family Physicians IRB.

Design details: Price DW, Wang T, O'Neill TR, et al. Acad Med. 2025;100:94–102.

Participants and Analytic Sample

16,751 physicians completed Q0, Q6, and Q10 (63.8% of the original Q0 cohort)



Restricting to responses that moved from incorrect to correct isolates confidence change that accompanies learning.

Characteristic	Category	n	%
Study group	Control / Groups 1–5	2,744–2,896 each	16.4–17.3
Sex	Male	8,763	52.3
	Female	7,970	47.6
Race	White	12,093	72.2
	Asian	2,495	14.9
	Black or African American	769	4.6
Age	<40 y	3,338	19.9
	50–59 y (largest group)	4,973	29.7
	70+ y	676	4.0
Practice	Direct patient care	12,597	75.2
	Urban community	9,090	54.3
	6–20 providers	4,681	27.9

Table 1 in Wang et al. Balanced group sizes (~16–17% each) support comparability across arms. Percentages include missing data in the denominator.

Analytic Approach

Within-person design: each physician serves as their own control

1

Define learning responses

Questions answered incorrectly at Q0 and correctly at Q6 (repeated items) or at Q10 (clone items). Confidence change = follow-up rating – baseline rating.

2

Test the change

One-sample t tests of the within-person confidence difference for each of the six study groups, separately for repeated (Q0→Q6) and clone (Q0→Q10) items.

3

Size the effect

Cohen d with 95% confidence intervals, because with $N \approx 2,700$ per group nearly any difference reaches $P < .05$. Benchmarks: 0.2 small, 0.5 moderate, 0.8 large.

4

Adjust for clinician characteristics

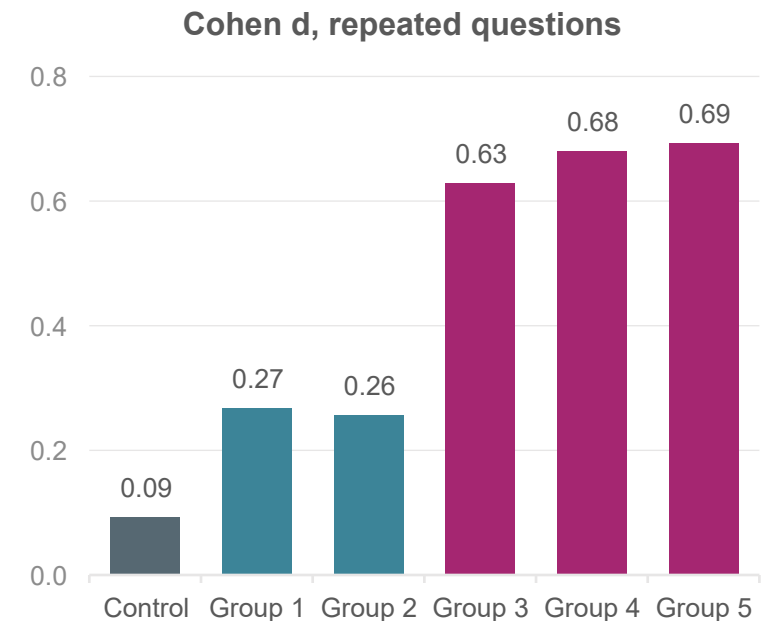
Multivariable regression of Q6 confidence on study group, controlling sex, age, race, ethnicity, patient-care role, practice size, faculty status, exam score, and community type.

Confidence treated as continuous (6-point ordinal scale). Effect-size benchmarks follow Cohen (2013).

Results 1: Repeated Questions (Q0 → Q6)

Two-time repetition produced large effects; the control group barely moved (Table 2)

Group	n	Q0	Q6	Δ	P	Cohen d (95% CI)
Control	2,722	2.819	2.868	0.048	.017	0.092 (0.017, 0.167)
Group 1	2,741	2.799	2.931	0.132	<.001	0.267 (0.192, 0.342)
Group 2	2,669	2.834	2.964	0.130	<.001	0.256 (0.180, 0.333)
Group 3	2,737	2.805	3.120	0.315	<.001	0.628 (0.551, 0.705)
Group 4	2,721	2.837	3.192	0.355	<.001	0.679 (0.602, 0.757)
Group 5	2,706	2.814	3.151	0.338	<.001	0.692 (0.614, 0.769)



Key finding

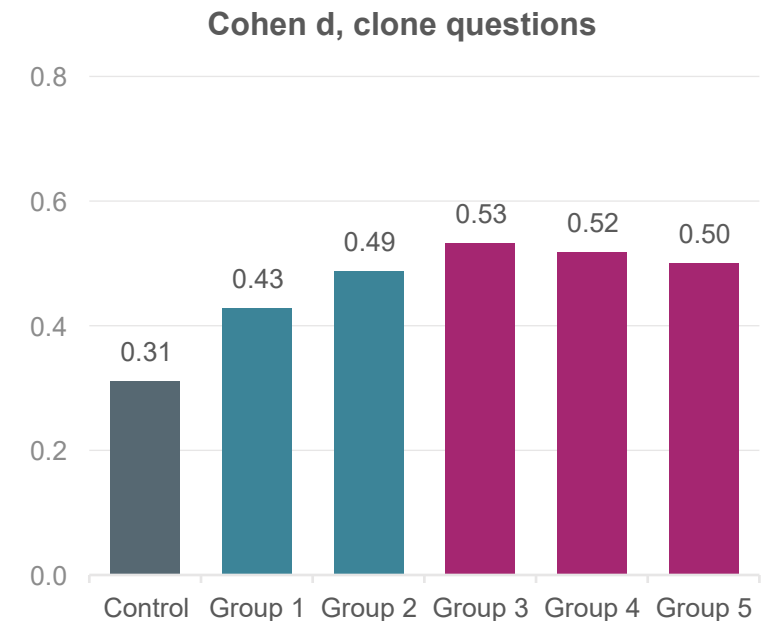
Baseline confidence was ≈ 2.8 in every arm. Control: negligible gain ($\Delta = 0.05$, $d = 0.09$). One repetition (Groups 1–2): moderate gain ($d \approx 0.26$). Two repetitions (Groups 3–5): mean gain of 0.32–0.36 points and large effects ($d = 0.63$ – 0.69).

Table 2 in Wang et al. Analysis restricted to items answered incorrectly at Q0 and correctly at Q6. One-sample t tests; Cohen d with 95% CI.

Results 2: Clone Questions (Q0 → Q10)

Gains transferred to novel items in every arm, with smaller magnitude than for repeated items

Group	n	Q0	Q10	Δ	P	Cohen d (95% CI)
Control	2,837	2.826	2.987	0.160	<.001	0.310 (0.236, 0.384)
Group 1	2,776	2.806	3.020	0.214	<.001	0.427 (0.352, 0.503)
Group 2	2,687	2.834	3.090	0.256	<.001	0.487 (0.410, 0.564)
Group 3	2,748	2.808	3.077	0.269	<.001	0.532 (0.456, 0.608)
Group 4	2,726	2.839	3.105	0.266	<.001	0.517 (0.441, 0.594)
Group 5	2,713	2.821	3.078	0.257	<.001	0.500 (0.424, 0.577)



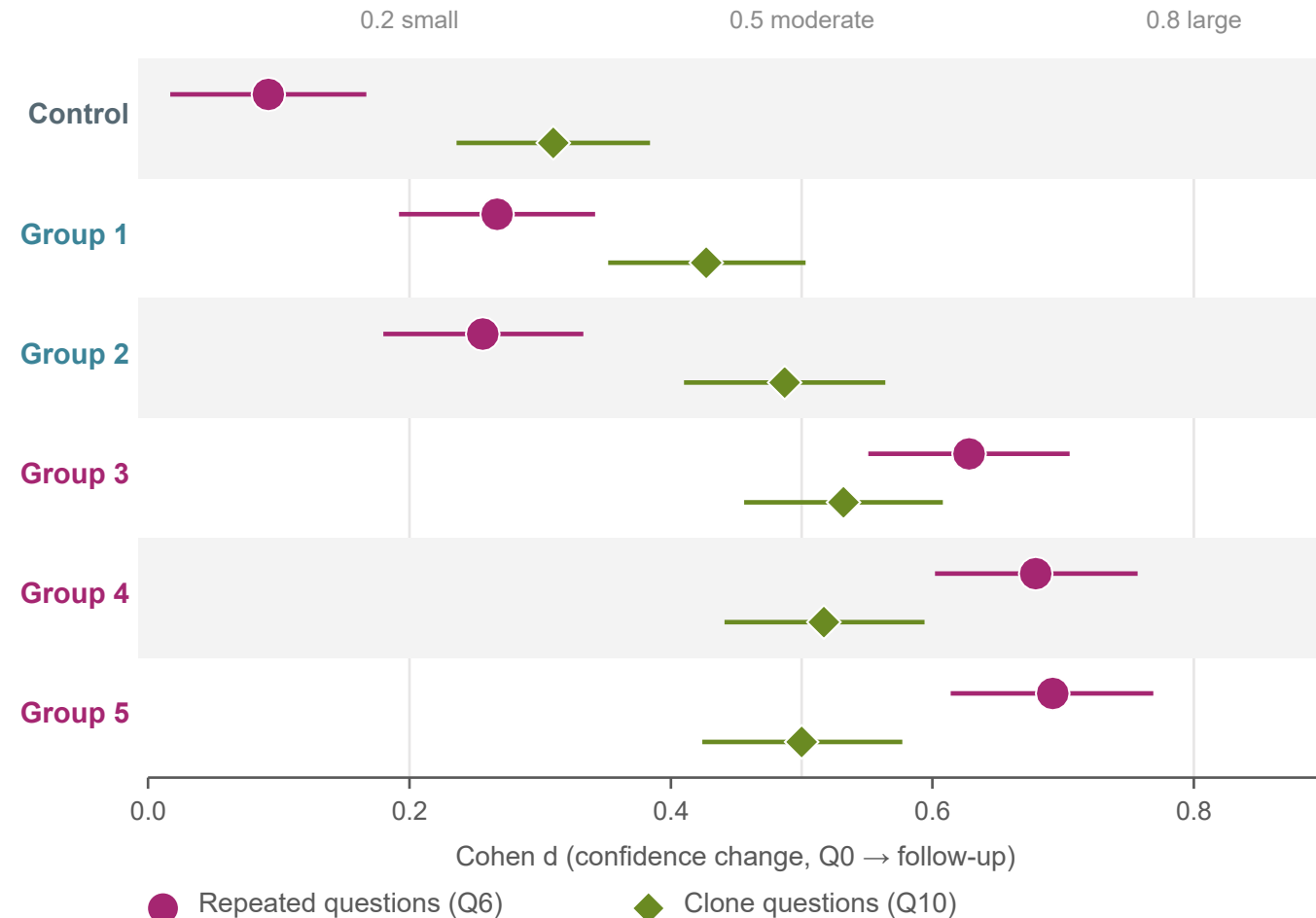
Key finding

All groups gained confidence on clone items four quarters after the repetition period ended. Control: small effect ($d = 0.31$). One repetition: moderate ($d = 0.43$ – 0.49). Two repetitions: moderate ($d = 0.50$ – 0.53), below the 0.63 – 0.69 seen for repeated items at Q6.

Table 3 in Wang et al. Analysis restricted to items answered incorrectly at Q0 and correctly at Q10. One-sample t tests; Cohen d with 95% CI.

Dose-Response and “Intermediate Confidence”

Effect sizes (Cohen d, 95% CI) for repeated and clone questions, by study group



Dose-response

Confidence gains rise stepwise with repetition frequency: control < one repetition < two repetitions, for both item types.

Intermediate-confidence effect

Clone-item effects ($d \approx 0.50$ – 0.53) fall between twice-repeated (0.63 – 0.69) and once-repeated (0.26 – 0.27) items.

Physicians grow confident after seeing an item twice, then temper that confidence when applying the concept in a new clinical context, bridging learning and transfer.

Results 3: Group Effects Persist After Adjustment

Multivariable model of confidence on repeated items at Q6, selected predictors (N = 16,751)

Predictor	Comparison	β (95% CI)	P
Study group	Group 1 vs control	0.05 (0.01, 0.09)	.012
	Group 2 vs control	0.09 (0.05, 0.13)	<.001
	Group 3 vs control	0.11 (0.07, 0.15)	<.001
	Group 4 vs control	0.17 (0.13, 0.21)	<.001
	Group 5 vs control	0.16 (0.12, 0.20)	<.001
Sex	Female vs male	-0.34 (-0.36, -0.31)	<.001
Age	60–69 y vs <40 y	-0.10 (-0.17, -0.03)	.004
	70+ y vs <40 y	-0.17 (-0.26, -0.08)	<.001
Exam score	>600 vs <400	-0.09 (-0.15, -0.03)	.003
Patient care	No direct care vs yes	-0.17 (-0.27, -0.07)	.001
Community	Isolated vs urban	-0.11 (-0.19, -0.04)	.004



Intervention effect is robust

Every intervention arm exceeds control after adjustment; two-repetition groups show the largest coefficients ($\beta = 0.11$ – 0.17).



Known demographic patterns replicate

Women and physicians aged 60+ report lower confidence, consistent with prior literature (Kinney 2021; Vajapey 2020).



Higher performers are more cautious

Physicians with higher certification exam scores rated confidence lower, in line with more careful self-assessment.

Table 4 in Wang et al. Model also adjusts for race, ethnicity, outpatient continuity care, practice size, and faculty status. Intercept = 3.16.

Discussion: What the Findings Mean

Spaced repetition improves not only what physicians know, but how well they know it

Metacognitive accuracy improves	Frequency matters	Format matters	Design strengthens inference
Among items physicians learned (wrong → right), confidence rose in step with correctness, so calibration improved rather than confidence alone.	A clear dose–response: marginal gains without repetition, moderate with one, largest with two. Deliberate repeated practice refines metacognitive judgment.	Repeated items produced larger gains than clone items. Familiarity reinforces metacognitive alignment more strongly than transfer tasks alone.	Randomization plus a within-person design limit confounding; the group effect survives adjustment for demographics known to shape confidence.

Consistent with theory and prior evidence: structured reflection calibrates confidence and competence (Gottlieb 2022); spaced learning improves retention in randomized trials (Kerfoot 2007); feedback and reflection drive self-regulated improvement (Watling & Ginsburg 2019).

Implications for Boards and CME Programs

Three practical steps that follow from the evidence

1

Repeat high-priority items, ideally twice

Two spaced repetitions boosted both knowledge retention (prior trial) and confidence calibration (this study), helping clinicians avoid overconfident mistakes.

2

Add a clone item a few quarters later

A rewritten scenario tests whether the concept transfers to a fresh clinical context and whether confidence stays appropriately calibrated.

3

Build calibration into learner dashboards

Visualize confidence against correctness over time, flag confident-incorrect responses, and embed reflective prompts at key intervals.

Relevance across ABMS boards: longitudinal assessment platforms already collect item-level confidence. Spaced repetition and clone items are low-cost design levers that can turn those ratings into a metacognitive learning signal.

Limitations

What this analysis can and cannot tell us

1

Learning responses only

Only wrong → right responses were analyzed. Persistent “confident incorrectness” and forgetting (right → wrong) were not examined because of small counts.

3

Retention and selection

Only physicians who remained through Q10 (63.8%) were included; findings may not extend to non-participants in longitudinal formative assessment.

5

Single specialty

Family medicine only. Because the mechanism rests on general adult-learning principles, similar patterns are hypothesized in other specialties.

2

Subjective ratings

“Somewhat confident” differs across people and contexts; stakes alter confidence for items of equal difficulty. Randomization and large samples make systematic group differences unlikely.

4

Cohort-level pooling

Pooled estimates mask individual differences; interviews with high-change individuals could show how clinicians reflect on their own learning.

6

Practical magnitude

The real-world meaning of a 0.2–0.3 point shift on a 6-point scale is uncertain; larger shifts may need reinforcement in actual practice.

Conclusions and Lessons for Practice

Supports both knowledge retention and the metacognitive accuracy needed for performance

Spaced repetition enhances metacognitive accuracy

Repeating questions, especially with a two-time spaced strategy, significantly increases confidence in correct responses, supporting more accurate self-assessment.

Gains depend on item type and repetition frequency

Repeated items yield larger confidence gains than clone items; two repetitions outperform one. Familiarity reinforces calibration more than transfer alone.

Demographic differences persist but do not blunt the effect

Sex, age, and practice setting shape confidence levels, yet the benefit of spaced repetition holds across diverse clinician populations.

Future research: link metacognitive change to downstream performance outcomes, examine confident-incorrect responses, and test generalizability across specialties and learner populations.

Thank you!

Ting Wang, PhD | twang@abpeds.org

Wang T, Morgan ZJ, Bazemore A, Newton WP, Price DW. J Contin Educ Health Prof. 2026;46(1):16-22.
doi:10.1097/CEH.0000000000000615



THE AMERICAN BOARD *of* PEDIATRICS